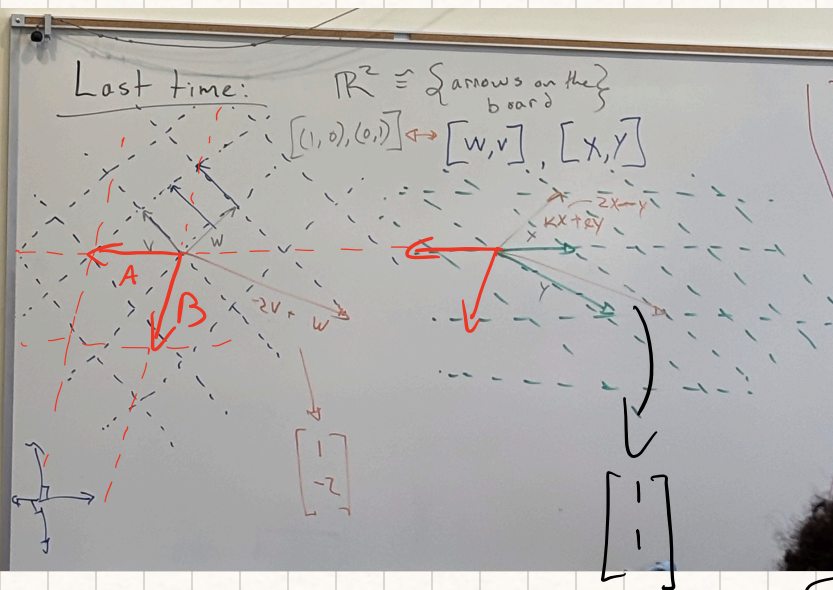




There are questions we can ask about vectors in these 2 spaces

basis dependent $\rightarrow$ have to agree on a basis

What are the coordinates of a vector?

$A = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

basis independent $\rightarrow$ get to choose a convenient basis

Are a set of vectors linearly independent or not

Is one vector in the span of some set of vector

$F: \{\text{arrows}\} \rightarrow V$ is F invertible?
 $F(1, -1) = k \quad F(1, 1) = l \quad [(1, -1), (1, 1)]$

are the columns of F linearly independent - injectivity

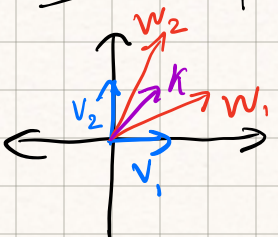
Thinking of vector spaces geometrically

$V \cong \mathbb{R}^n$

$V \cong \mathbb{F}^n$

$\mathbb{C}^n - \mathbb{C}$ is easy to visualize $\mathbb{C}^2$ is tricky

Ex: $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, linear



$F(v_1) = w_1 \quad F(v_2) = w_2$

We can write all vectors in $\mathbb{R}^2$ as a linear combination of $[v_1, v_2]$

$$v_1 \leftrightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad v_2 \leftrightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad w_1 \leftrightarrow \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad w_2 \leftrightarrow \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

So the matrix for F is $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

$$k = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad F(k) = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3k \quad \leftarrow \begin{array}{l} \text{our eigenvector} \\ \text{our eigenvalue} \end{array}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad F(\text{sam}) = \text{sam} \quad \leftarrow \begin{array}{l} \text{another eigenvector} \\ \text{other eigenvalue is 1} \end{array}$$

$[k, \text{sam}]$ is a basis

$$F \text{ is } \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

Steps:

1) gave 2 bases for $\mathbb{R}^2$ $[v_1, v_2]$ $[w_1, w_2]$

2) defined a linear function F in terms of this basis

$$F(v_1) = w_1 \quad F(v_2) = w_2$$

3) expressed all of this in coords relative to v basis

4) found 2 vectors satisfying $Fx = \lambda x$

$$Fk = 3k \quad F(\text{sam}) = 1(\text{sam})$$

$[k, \text{sam}]$ is linearly independent and spanning $\Rightarrow$ basis

The matrix for F w/ $[k, \text{sam}]$ basis is nice i.e. diagonal

Def: let $H: V \rightarrow V$ be linear if there is λ and $x \in V$

s.t. $Hx = \lambda x$ we call x an eigenvector of H and λ an eigenvalue of H